LESSON 08 · CHEAT SHEET

NumPy

MAKING ARRAYS

```
import numpy as np
a = np.array([1.5, 2.0, 3.5])       # from a list
a = np.array(rewards)               # list -> column
a.shape    (400,)     a.dtype   float64
a.size     400        len(a)    400
np.zeros(5)   np.arange(1, 13)   np.linspace(0, 1, 5)
```

COLUMN MATHS (NO LOOPS)

```
a / 2           # every element
a * 1.10        # every element
a / days        # row by row, same length
a - a.mean()    # difference from the mean
a.round(2)      # round every element
np.where(a > 1000, "rich", "lean")  # IF()
np.maximum(a, 25)     # floor every element
```

SUMMARIES

`a.sum()`	SUM
`a.mean()`	AVERAGE
`a.min()` / `a.max()`	MIN / MAX
`np.median(a)`	MEDIAN
`a.std()`	STDEV.P
`np.percentile(a, 75)`	PERCENTILE(a, 0.75)
`np.percentile(a, [25, 50, 75])`	three at once
`a.argmax()`	position of the max

MASKS: AUTOFILTER IN ONE LINE

```
big = a > 1000       # array of True/False
big.sum()            # how many True (COUNTIF)
big.mean()           # share of True
a[big]               # only those rows
a[big].mean()        # AVERAGEIF
a[monsters == "Wyrm"]  # mask, other column

(a > 1) & (b == 2)   # and   (brackets!)
(a > 1) | (b == 2)   # or
~big                 # not
```

INDEXING & SLICING

```
a[0]    a[-1]    a[:5]    a[10:20]
a[[0, 3, 7]]            # pick positions
a[big] = 0              # assign through a mask
b = a.copy()            # a real copy
```

2-D: A WHOLE RANGE

```
m = np.array([[1, 2, 3],
              [4, 5, 6]])
m.shape        (2, 3)  # rows, columns
m[1, 2]        6       # row 1, column 2
m[0]           [1 2 3] # whole row
m[:, 1]        [2 5]   # whole column
m.sum()        21      # everything
m.sum(axis=0)  [5 7 9] # down: one per column
m.sum(axis=1)  [6 15]  # along: one per row
```

EXCEL → NUMPY

drag `=B2/12`	`a / 12`
`COUNTIF`	`(a > 1000).sum()`
`AVERAGEIF`	`a[mask].mean()`
`IF()` dragged	`np.where(m, x, y)`
`QUARTILE(a, 1)`	`np.percentile(a, 25)`
SUM under column	`m.sum(axis=0)`

GOTCHAS

- `and` / `or` crash on arrays: use `&` / `|` with brackets
- `a.mean` without `()` is the method, not the number
- `np.percentile(a, 0.75)` is (nearly) the minimum. Use 75
- Arrays have one type: `np.array([1, "a"])` turns everything into text
- `axis=0` is down, `axis=1` is along. Everyone swaps them once